

Fig. 8. Examples of long-instruction execution in Habitat. Each row pairs a natural-language instruction with the corresponding sequence of first-person observations, covering freely ordered subtasks and observation-dependent search or verification.

effective order, forbids visits to skipped branches, and checks spatial/reference constraints.

As shown in Table II, removing CPTG reduces multi-target SR to 64.84% and SPL to 61.85%, while increasing successful-task CT to 35.63 s. Without an explicit task graph, the system cannot reliably enforce precedence relations, activate observation-dependent branches, or distinguish required tasks from skipped alternatives. This leads to incorrect execution orders, unnecessary visits, and repeated recovery attempts, demonstrating that CPTG is the principal contributor to full-task correctness. Restoring CPTG with fixed viewpoints raises SR by 31.86 percentage points to 96.70%, increases SPL to 78.03%, and reduces CT to 11.65 s. These improvements show that explicit task dependencies and conditional transitions eliminate most logical execution failures and substantially reduce redundant motion. Replacing the independently selected fixed viewpoints with JTVP preserves the same 96.70% SR while further increasing SPL by 4.12 percentage points to 82.15% and reducing CT by 5.26 s to 6.39 s. Thus, CPTG primarily ensures correct execution of the complete instruction, whereas JTVP improves efficiency by jointly selecting task order and observation poses according to travel and terminal-view costs. The 30 timeout or infrastructure-failure outcomes of w/o CPTG remain in the 546-episode denominator.

D. Evaluation in Real-World Environments

Experimental platform. We conduct real-world experiments using a custom-built quadrotor equipped with an NVIDIA Jetson Orin NX (16 GB), a PX4 flight controller, a Livox Mid-360 LiDAR, and an Intel RealSense D435 depth camera. The D435 provides synchronized 640×480 RGB and depth observations for online verification and 3D projection. FAST-LIO [40] provides onboard localization, while LiDAR-based occupancy mapping supports real-time trajectory optimization. The UAV is constrained to a maximum linear velocity of 1.0 m/s, linear acceleration of 1.0 m/s^2, and angular velocity of 1.0 rad/s.

As illustrated in Fig. 1, LogicFly completes a four-task instruction end to end with a physical UAV in a real multi-room environment. After the laptop is not found at its instructed location, the UAV performs semantic recovery in the reading room, then verifies the laptop, toolbox, microwave state, and water bottle. The map panel provides a bird's-eye visualization of the executed route, while numbered views document the recovery and verification sequence.

Figure 9 summarizes real-world deployments in a confined home and two office areas, showing representative UAV flights and target detections.